

Paper 4A: Further Pure Mathematics 2

Topics	What students need to learn:		
		Content	Guidance
1 Groups	1.1	The Axioms of a group.	The terms 'binary operation, closure, associativity, identity and inverse'.
	1.2	Examples of groups. Cayley tables. Cyclic groups.	For example, symmetries of geometrical figures, non-singular matrices, integers modulo n with operation addition, and/or multiplication permutation groups.
	1.3	The order of a group and the order of an element. Subgroups.	
	1.4	Lagrange's theorem.	
	1.5	Isomorphism.	Isomorphisms will be restricted to groups that have a maximum order of 8.